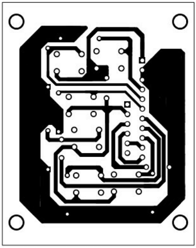

Fig. 5: Actual-size PCB layout of transmitter unit

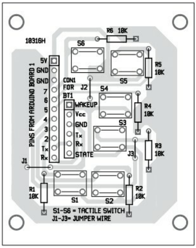

Fig. 6: Components layout for the PCB in Fig. 5

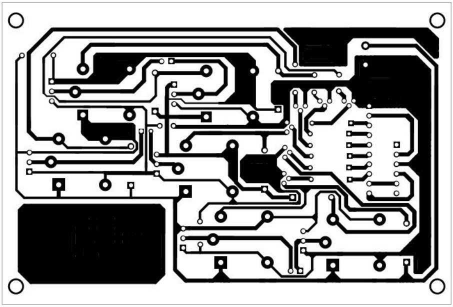

Fig. 7: Actual-size PCB layout of receiver unit

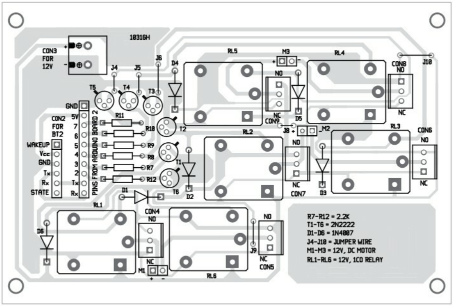

Fig. 8: Components layout for the PCB in Fig. 7

Construction and testing

An actual-size PCB layout of the transmitter unit is shown in Fig. 5 and its components layout in Fig. 6. Similarly, actual-size PCB layout of the receiver unit is shown in Fig. 7 and its components layout in Fig. 8.

Use two 50rpm DC motors for the wheels of the robot, one for right wheel and the other for left wheel. A castor wheel can be used as front wheel of the robot. Use another 50rpm DC motor for the object-lifting arm.

Connect 9V battery to both transmitter and receiver units. Connect 12V battery for the DC motors in the receiver unit.

Switch on both the transmitter and receiver units. If status LEDs of both BT1 and BT2 start blinking twice every second simultaneously, that means both devices are connected to each other. Now you can press any switch on the transmitter unit to control the robot. Switch S6 is used for lifting or cutting the object. Switches S1 through S5 are used for various movements of the robot. EFY

EFY Note
The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

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